

MAREEBA, Australia

WMO: 952860

Lat: 17.067S

Lon: 145.428E

Elev: 473

StdP: 95.77

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	12.3	3.5	5.1	19.9	5.6	6.0	18.3	9.3	19.4	8.5	20.0	2.9	140	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	9.5	33.2	21.5	31.9	21.7	30.9	21.6	24.6	28.5	24.2	27.8	23.8	27.3	4.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.7	25.6	23.3	19.2	25.3	23.0	18.8	25.1	77.7	28.7	75.9	27.8	74.3	27.3	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.8	7.1	8.5	36.3	1.5	1.3	7.4	37.3	6.6	38.0	5.7	38.8	4.7	39.7
			6.6	25.7	1.3	0.9	5.6	26.4	4.9	26.9	4.1	27.4	3.2	28.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	25.8	25.6	24.7	23.3	21.5	20.0	19.1	19.6	21.5	23.4	24.8	25.9	(6)
	DBStd	2.83	1.34	1.22	1.21	1.32	1.50	1.71	1.73	1.71	1.59	1.62	1.37	1.36	(7)
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD18.3	27	0	0	0	0	1	6	12	7	1	0	0	0	(9)
	CDD10.0	4713	490	436	456	399	356	298	283	297	346	416	444	492	(10)
	CDD18.3	1698	232	202	198	149	99	54	36	47	97	157	194	234	(11)
	CDH23.3	10598	1665	1366	1157	703	322	139	96	208	606	1083	1422	1832	(12)
(13)	CDH26.7	3003	538	400	271	110	27	8	4	18	130	331	483	682	(13)
(14)	Wind	WSAvg	3.4	2.7	2.6	3.3	3.9	3.9	4.0	3.7	3.3	3.3	3.1	2.9	(14)
(15) Precipitation	PrecAvg	954	206	239	187	46	27	24	15	13	10	22	55	111	(15)
	PrecMax	1603	523	643	450	255	85	81	41	90	33	106	204	362	(16)
	PrecMin	537	59	30	24	0	0	1	0	0	0	0	0	11	(17)
	PrecStd	285	109	154	121	56	23	18	12	19	9	24	58	81	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.4	33.4	31.8	30.2	28.8	27.7	27.1	28.6	31.4	33.7	34.5	34.7	(19)
		MCWB	22.0	22.5	22.3	21.1	20.1	19.6	18.3	18.4	17.9	19.2	20.6	21.4	(20)
	2%	DB	32.1	31.6	30.3	28.9	27.3	26.1	25.5	26.9	29.5	31.4	32.0	32.8	(21)
		MCWB	22.6	23.0	22.5	21.2	19.8	19.1	18.0	17.8	18.2	19.6	20.5	21.9	(22)
	5%	DB	30.8	30.4	29.3	28.0	26.2	24.9	24.3	25.6	28.2	29.9	30.6	31.5	(23)
		MCWB	22.5	22.9	22.2	20.9	19.5	18.6	17.5	17.5	18.3	19.7	20.8	21.9	(24)
	10%	DB	29.6	29.2	28.2	27.0	25.2	23.8	23.2	24.4	26.8	28.5	29.4	30.2	(25)
		MCWB	22.5	22.9	22.1	20.7	19.2	18.1	17.1	17.1	18.1	19.6	20.7	21.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.2	25.1	25.0	23.9	22.1	21.3	20.6	21.1	22.0	23.0	24.1	24.9	(27)
		MCDB	29.3	29.3	28.8	27.8	26.1	25.0	23.9	25.7	27.4	28.3	29.0	28.7	(28)
	2%	WB	24.5	24.6	24.1	23.0	21.3	20.3	19.6	19.7	20.9	22.1	23.1	24.1	(29)
		MCDB	28.2	28.4	27.5	26.5	24.9	23.9	23.1	24.2	25.8	27.2	27.7	28.2	(30)
	5%	WB	24.0	24.1	23.6	22.4	20.7	19.6	18.8	18.9	20.1	21.4	22.5	23.6	(31)
		MCDB	27.5	27.5	26.8	25.7	24.2	23.1	22.4	23.4	24.9	26.4	27.1	27.7	(32)
	10%	WB	23.6	23.7	23.2	21.7	20.1	18.9	18.1	18.1	19.3	20.7	21.9	23.2	(33)
		MCDB	26.9	26.8	26.2	25.0	23.7	22.6	21.9	22.5	24.2	25.7	26.6	27.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	7.7	7.4	7.9	8.1	8.5	9.1	10.2	11.5	11.1	10.4	9.5	(35)
		MCDBR	10.6	9.9	9.4	9.5	9.5	9.9	10.4	12.1	13.6	13.3	12.3	11.8	(36)
		MCWBR	3.0	3.0	3.1	3.2	3.5	4.0	4.1	4.4	4.5	3.9	3.4	3.2	(37)
	5% WB	MCDBR	8.0	7.7	7.3	7.2	7.6	8.1	7.8	10.0	11.0	10.6	10.2	9.1	(38)
		MCWBR	2.9	2.8	3.0	3.0	3.3	3.7	3.5	4.1	4.3	3.7	3.3	3.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.377	0.375	0.363	0.352	0.338	0.330	0.315	0.326	0.353	0.371	0.377	0.385	(40)
	taud		2.577	2.591	2.612	2.617	2.618	2.621	2.655	2.601	2.501	2.476	2.499	2.512	(41)
	Ebn at Noon		967	959	949	924	903	893	919	937	944	952	960	959	(42)
	Edh at Noon		107	104	99	92	86	83	82	92	109	116	115	114	(43)
(44) All-Sky Solar Radiation	RadAvg		5.79	5.57	5.21	4.78	4.22	3.82	4.20	5.08	6.10	6.59	6.71	6.42	(44)
	RadStd		0.45	0.65	0.46	0.48	0.37	0.37	0.31	0.42	0.36	0.48	0.55	0.73	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page